# Introduction to Git & GitHub

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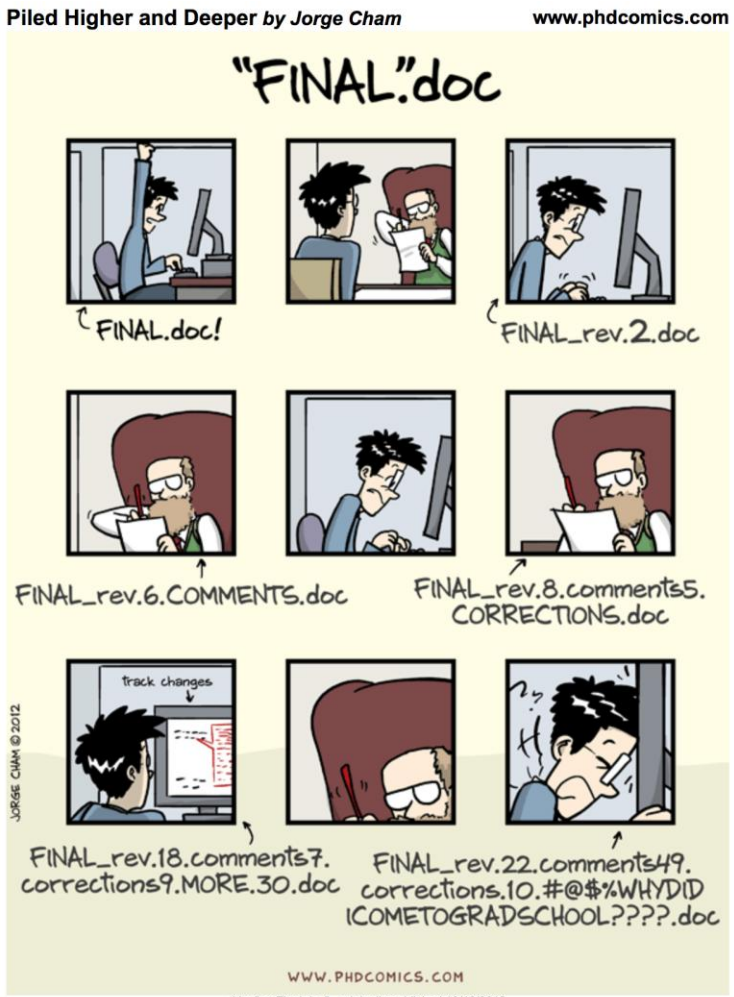
SAMS Session 1

31st March 2026

# Why use version control?

- How many people have directories that include files like this?
  - `File_16042025.doc, File_30052025.doc, …`
  - `File_old.doc, File_new.doc`
  - `File.doc.orig`
  - `File.doc, File_test.doc`

- Version control software allows you to:
  - Back up your code.
  - Keep track of changes.
  - Share your code with others.
  - Develop code with collaborators.

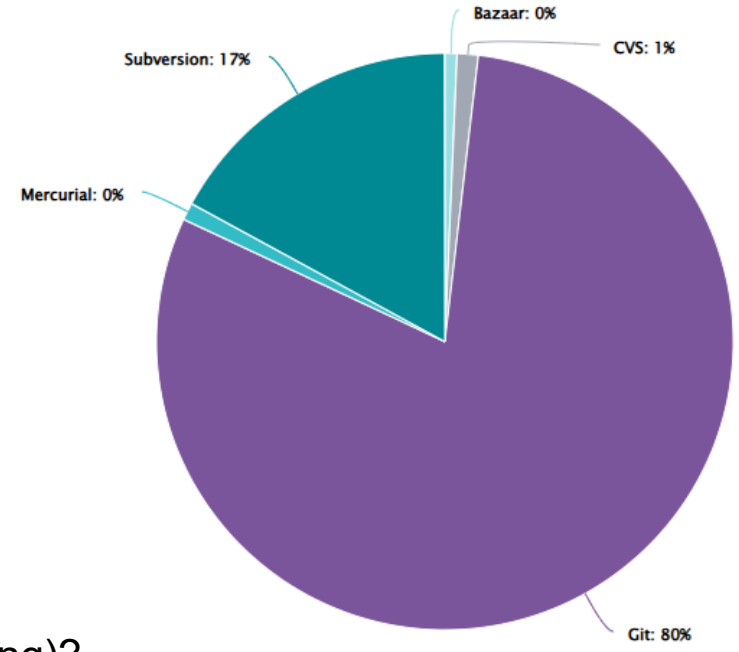


# Changes are documented

- As code is backed up, can be confident in testing new ideas.
- Version control software ensures that changes are attributable to individuals.
  - If the code breaks, you know why!
- Code can be shared along with the documentation and file history.
  - Documentation can explain how and why the code should be used.
  - Can create fixed versions/releases, along with DOIs.
    - Sharing now often required for publication.

- All the above provides “quality assurance” in the final product.

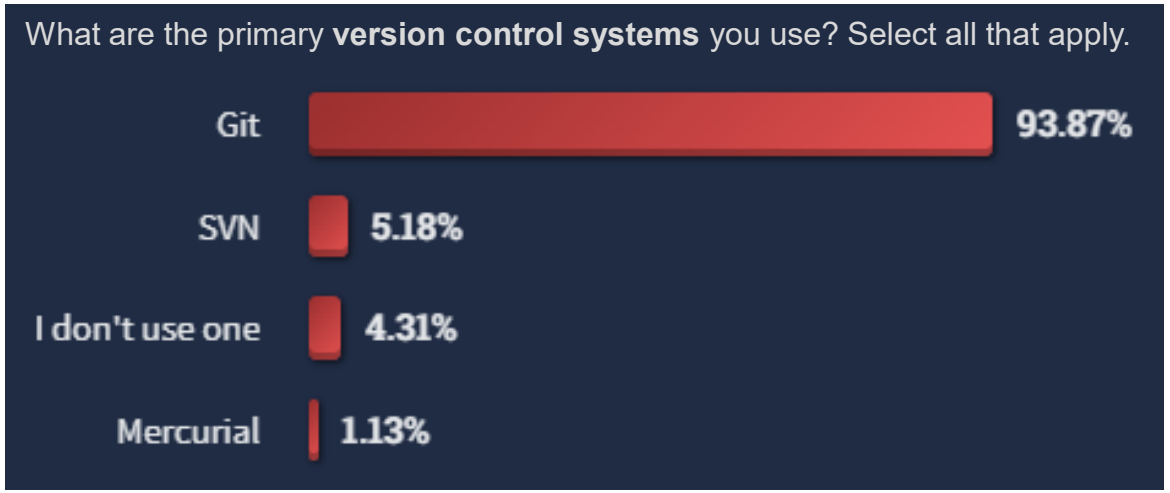
# Version control software



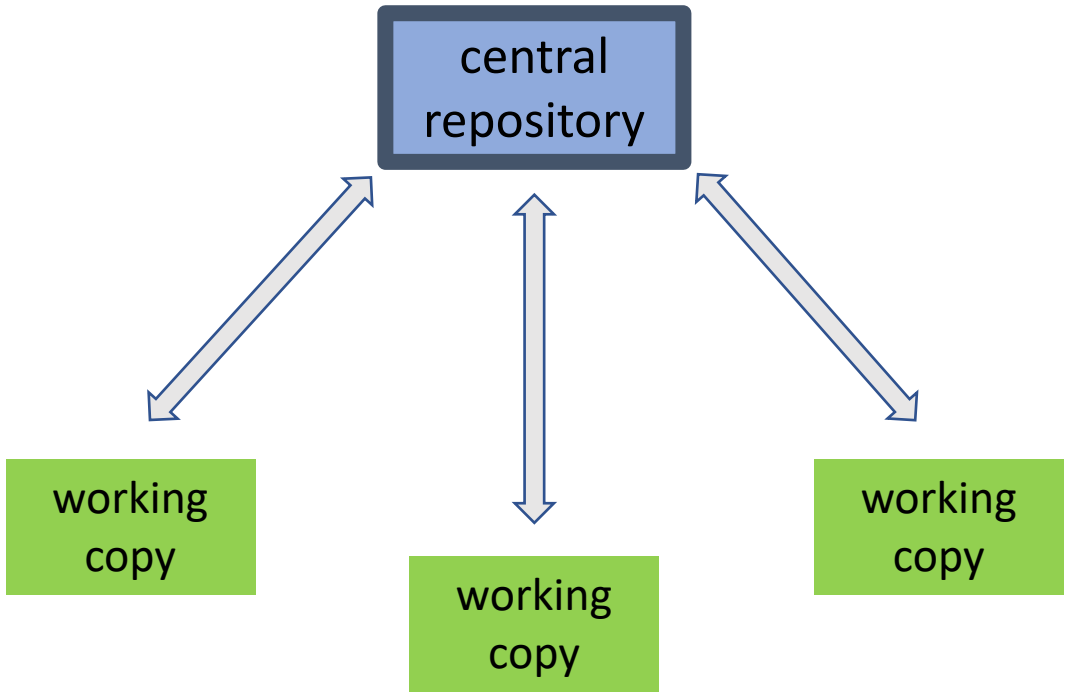

- Many options out there…
  - https://en.wikipedia.org/wiki/List_of_version-control_software
  - All are *agnostic* in terms of the programming language
- Client-server model :: all users share a single code repository.
  - e.g., SVN (subversion)
    - Legacy choice, depending on when project started (many now switching)?
- Distributed model :: all users have their own local repositories. Changes can be shared/merged as a separate step.
  - e.g., **Git**
    - What we're using today!
    - Well supported, large user community, e.g.,

    From Stack Overflow user survey (2022),

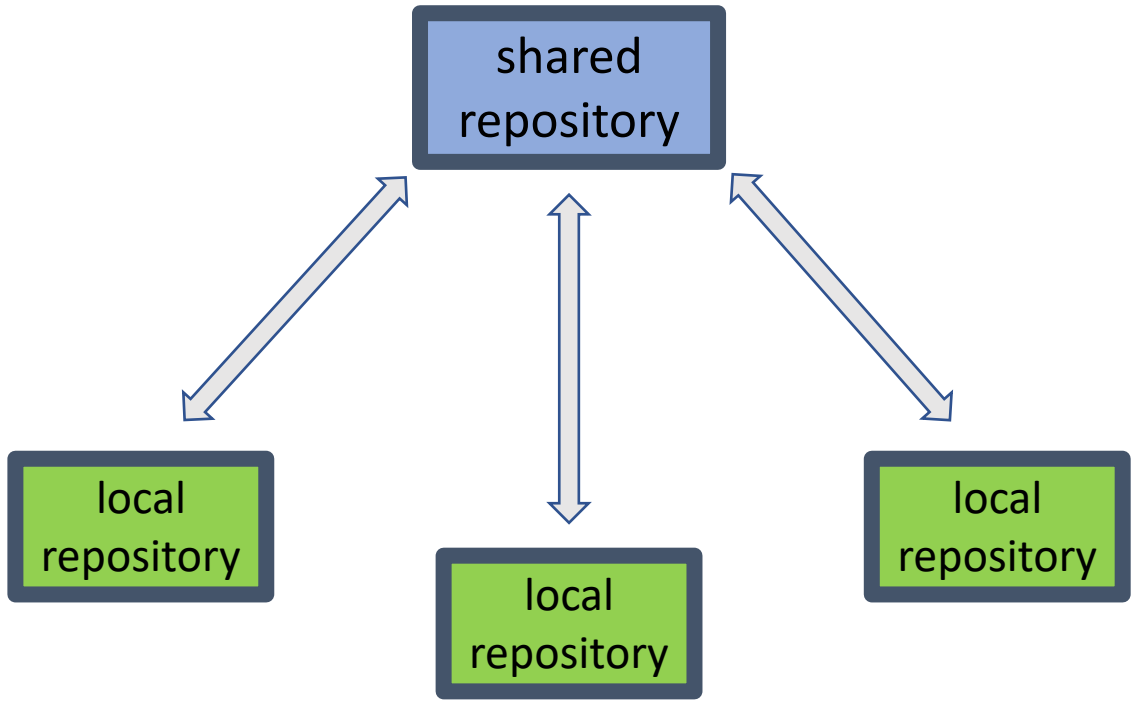
    *"No other technology is as widely used as Git."*



# Central vs Distributed workflow?




Centralised/client-server model (e.g., SVN)




Distributed model (e.g., Git)

# So, what are Git & GitHub?

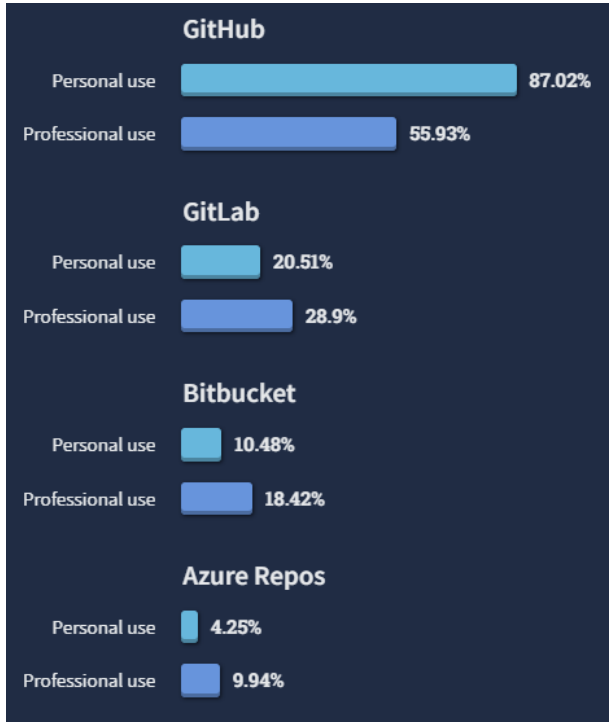
NB. These are <u>two different things</u>

- Git = version control software.
- GitHub = web-based repository hosting service*.

i.e. Git is the software behind the GitHub web service.

You can use Git without GitHub. However, GitHub provides useful tools (especially for sharing your code).

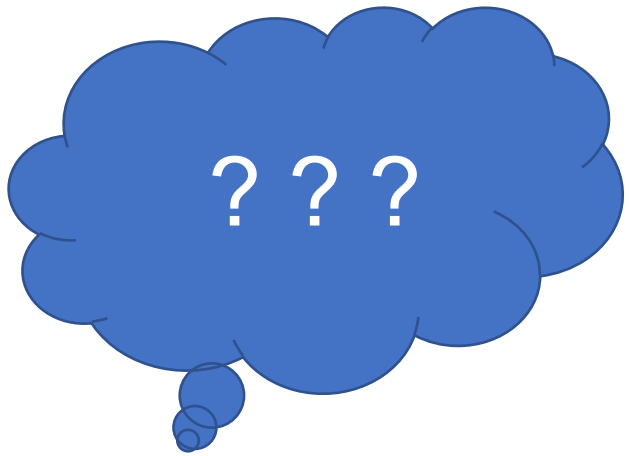
* Other providers exist, but GitHub is the most popular, e.g.:



# Useful resources

Git and GitHub are widely used, so there are lots of useful websites, training videos, online courses etc. e.g.,

- GitHub guides & help:
  - https://guides.github.com/
- Git guides:
  - https://git-scm.com/doc
- Git cheat sheet:
  - https://education.github.com/git-cheat-sheet-education.pdf
- E-books:
  - https://www.git-tower.com/learn/git/ebook/
- Webinars:
  - https://www.youtube.com/watch?v=v3Y8c2KMay8
  - https://www.youtube.com/watch?v=ShH1g4I9A54
- Internet searching!



# Using Git

- Various options depending on preference:
  - **Command line e.g., Git for windows** (bash shell)
    - Many other command line interfaces available e.g., HPC, Linux or Mac terminals.
  - GUI e.g., GitHub desktop
  - IDEs, e.g., Rstudio or pycharm (some built-in functionality, similar to GUI)
  - *[NB. Admin rights not needed to install Git]*

- Whatever option you choose, all the above follow the same basic workflow.
  - Can choose whatever works for you e.g., are you more comfortable with a GUI or command line?
  - Will initially take some practice, but using git can (should) become a regular activity.

https://xkcd.com/1597/

# Why are we using the command line?

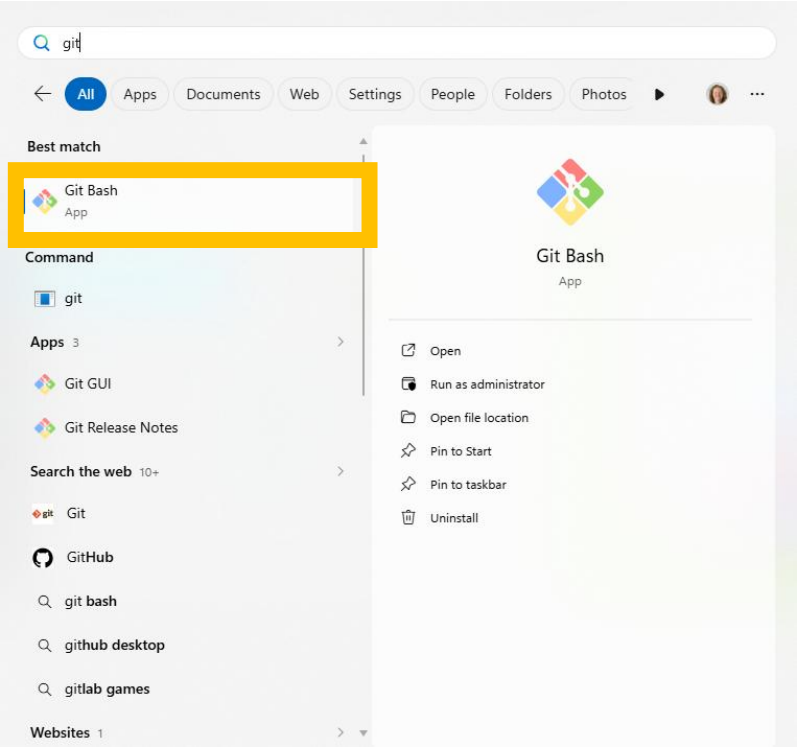
- **Clearly shows the commands being used at each step in the workflow.**
  - Command name matches exactly what is used on the command line.

- Can apply the workflow to a chosen GUI or IDE at later date (e.g., GitHub Desktop, Rstudio, pycharm).

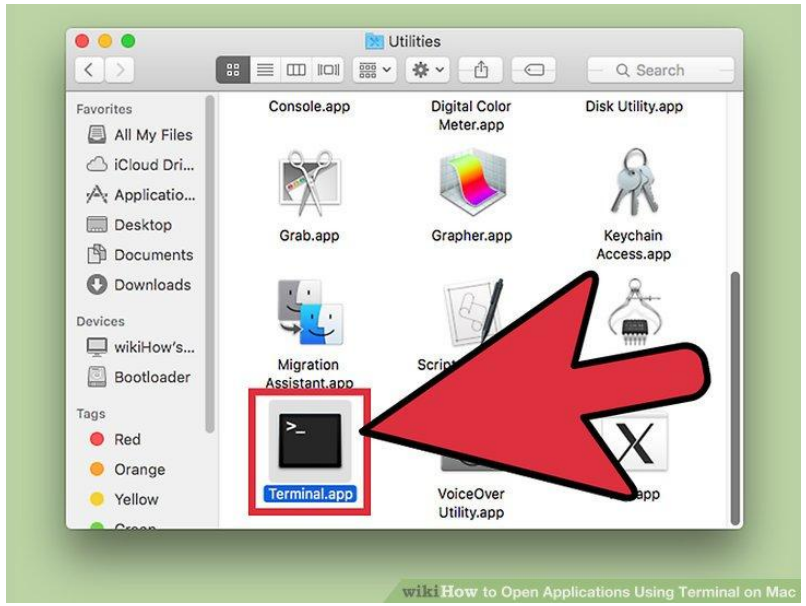
- Not all commands are available in every GUI, so always have the option of moving back to the command line.

# Where to find the command line?

- Windows
  - Open Git bash

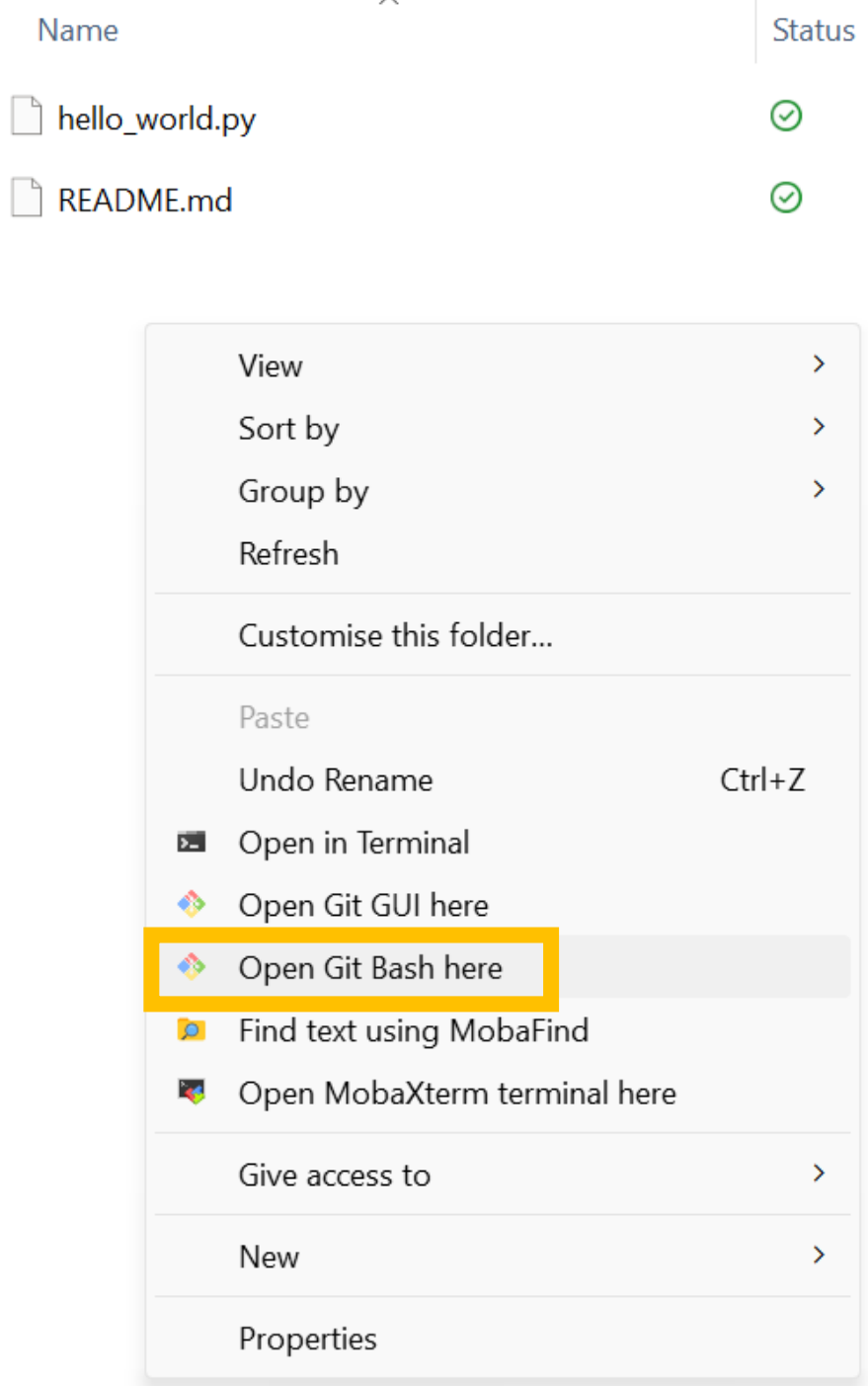



- MacOS
  - Open Terminal from
    - Applications -> Utilities



# Tip for Windows users

- Useful short cut to avoid navigating to working directory.
  - Right click within windows explorer
    - [Show more options]
  - <u>Open Git Bash here</u>



# Basic Git workflow

- Create a local repository (code directory).
- Add/edit a file.
- Commit your changes (i.e. document changes).
- Then repeat!

Optional (*but recommended*):

- Push changes to remote repository on GitHub (back-up/sharing).

# Creating a repository…

I have a folder or directory, how do I start tracking my changes?

```
JG@SAMS MINGW64 ~/OneDrive - SAMS/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive - SAMS/GitHub/Repos/hello_world
$
```

```
JG@SAMS MINGW64 ~/OneDrive - SAMS/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive - SAMS/GitHub/Repos/hello_world
$ git init
```

**git init**

:: turn current directory into git repository.

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ git init
Initialized empty Git repository in C:/Users/…/OneDrive/GitHub/Repos/hello_world/.git/

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ git init
Initialized empty Git repository in C:/Users/…/OneDrive/GitHub/Repos/hello_world/.git/

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
```

**git status**

:: show which files have been tracked or modified.

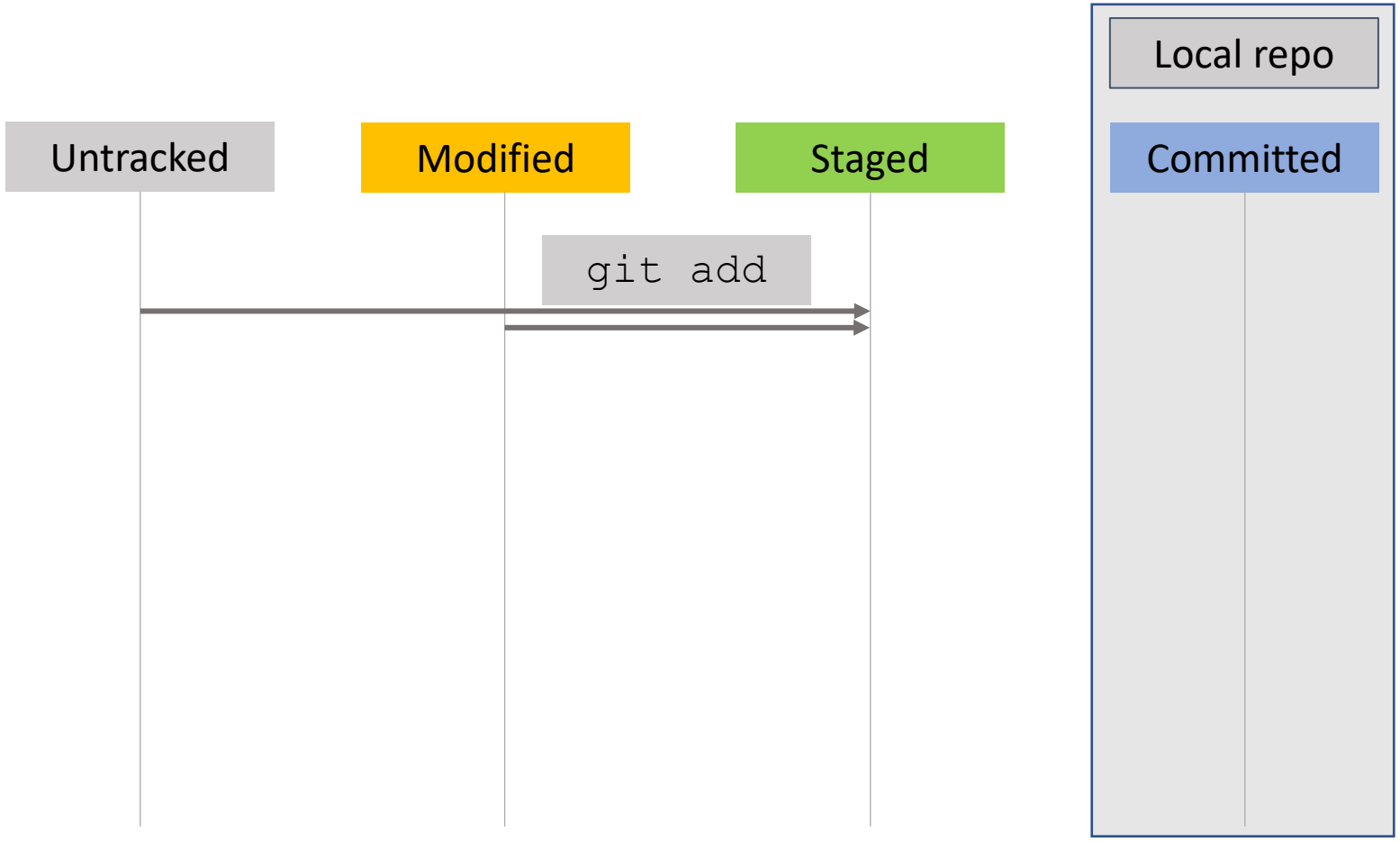
```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ git init
Initialized empty Git repository in C:/Users/…/OneDrive/GitHub/Repos/hello_world/.git/

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello_world.py

nothing added to commit but untracked files present (use "git add" to track)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

# The “Staging area”

- Git doesn’t have to track all files in a repository.

- There are three “states” of tracked files in a Git repository.
  - Working directory :: current version of files, containing any modifications.
  - Staging area :: snapshot of files/modifications that you plan to commit.
    Note: the Staging area is also often called the `index`.
  - Repository/Git directory :: contains the commit history for all files in the repository.

- You can check the state of each file at any point using “`git status`”


Local repo
Untracked
Modified
Staged
Committed
git add

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ git init
Initialized empty Git repository in C:/Users/…/OneDrive/GitHub/Repos/hello_world/.git/

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello_world.py

nothing added to commit but untracked files present (use "git add" to track)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add hello_world.py
```

**git add <file>**

:: stage the file for commit i.e. track changes.

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ ls
hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world
$ git init
Initialized empty Git repository in C:/Users/…/OneDrive/GitHub/Repos/hello_world/.git/

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello_world.py

nothing added to commit but untracked files present (use "git add" to track)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add hello_world.py
warning: in the working copy of 'hello_world.py', LF will be replaced by CRLF the next time Git
touches it

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add hello_world.py
warning: in the working copy of 'hello_world.py', LF will be replaced by CRLF the next time Git
touches it

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello_world.py


JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
```

**git commit**

:: confirm changes with message/explanation.

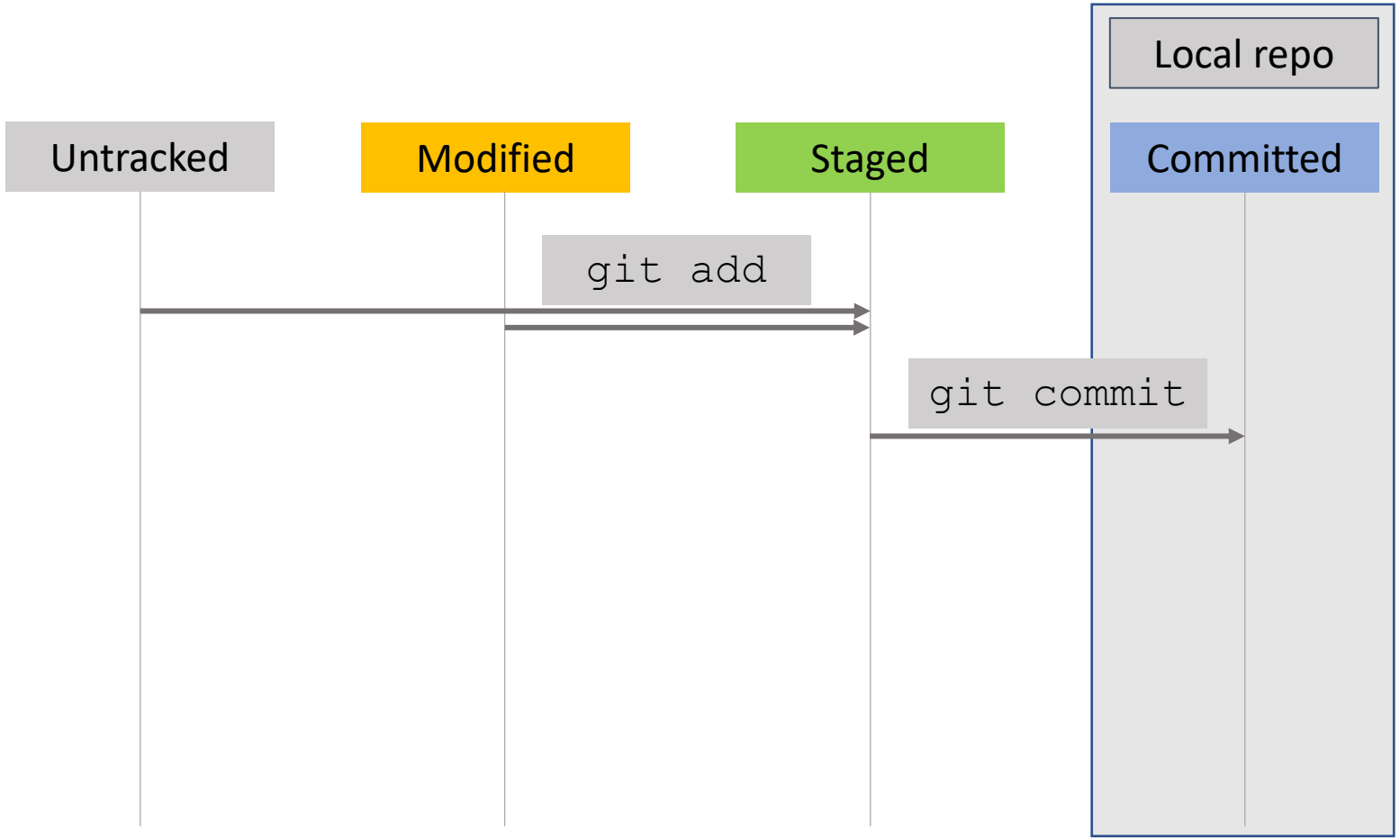
*COMMIT_EDITMSG - Notepad

File Edit Format View Help

```
Adding first file to repository
# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
#
# Initial commit
#
# Changes to be committed:
#       new file:   hello_world.py
#
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
hint: Waiting for your editor to close the file... unix2dos: converting file
C:/Users/sa04jg/OneDrive - SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to DOS format...
dos2unix: converting file C:/Users/sa04jg/OneDrive -
SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to Unix format...
[main (root-commit) d63f899] Adding first file to repository
 1 file changed, 9 insertions(+)
 create mode 100644 hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
hint: Waiting for your editor to close the file... unix2dos: converting file
C:/Users/sa04jg/OneDrive - SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to DOS format...
dos2unix: converting file C:/Users/sa04jg/OneDrive -
SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to Unix format...
[main (root-commit) d63f899] Adding first file to repository
 1 file changed, 9 insertions(+)
 create mode 100644 hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
nothing to commit, working tree clean

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```


Local repo
Untracked
Modified
Staged
Committed
git add
git commit

# Committing changes

- Commit your change.
  - You will be prompted for a commit message.
  - Make a concise, meaningful description of the changes and why they were made.
  - Best practice is to stage only associated modifications in each commit.

- How often should you commit?
  - More often than you think - whenever you make a change that works?
  - The more often you commit, the easier it is to document (with short, meaningful messages), or roll-back if needed.
  - *Avoid committing untested or unfinished modifications!*

https://xkcd.com/1296/

```
git init              :: turn current directory into git repository.
git add <file>        :: stage the file for commit i.e., track changes.
git commit            :: confirm changes with message/explanation.

git status            :: show which files have been tracked or modified.
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls
README.md  hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add README.md

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
hint: Waiting for your editor to close the file...
[main e2c5dde] Adding README file
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
nothing to commit, working tree clean
```

Create a new file, add and commit …

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls
README.md  hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ vi hello_world.py
```

Edit files and make changes …

```
'''
My first code.

@author JGraham
created 31/3/26
'''

hello = 'Hello wold'
print(hello)
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls
README.md  hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ vi hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git diff
```

git diff <file>

:: show difference between current file and last commit.

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls
README.md  hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ vi hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git diff
warning: in the working copy of 'hello_world.py', LF will be replaced by CRLF the next time Git
touches it
diff --git a/hello_world.py b/hello_world.py
index 2e7c37e..e094d10 100644
--- a/hello_world.py
+++ b/hello_world.py
@@ -5,5 +5,5 @@ My first code.
 created 31/3/26
 '''

-hello = 'Hello wold'
+hello = 'Hello world'
 print(hello)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

git diff <file>

:: show difference between current file and last commit.

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ vi README.md

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git diff README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git
touches it
diff --git a/README.md b/README.md
index eb6d976..0d08e3c 100644
--- a/README.md
+++ b/README.md
@@ -1 +1 @@
-This is my first repository.
+This is Jenny's first repository.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ vi README.md

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
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index eb6d976..0d08e3c 100644
--- a/README.md
+++ b/README.md
@@ -1 +1 @@
-This is my first repository.
+This is Jenny's first repository.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md
        modified:   hello_world.py

no changes added to commit (use "git add" and/or "git commit -a")

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add hello_world.py
warning: in the working copy of 'hello_world.py', LF will be replaced by CRLF the next time Git
touches it

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   hello_world.py

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md


JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

Don’t have to add all files to commit.

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add hello_world.py
warning: in the working copy of 'hello_world.py', LF will be replaced by CRLF the next time Git
touches it

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   hello_world.py

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md


JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
hint: Waiting for your editor to close the file... unix2dos: converting file
C:/Users/sa04jg/OneDrive - SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to DOS format...
dos2unix: converting file C:/Users/sa04jg/OneDrive -
SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to Unix format...
[main 01b729d] Fixing typo
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

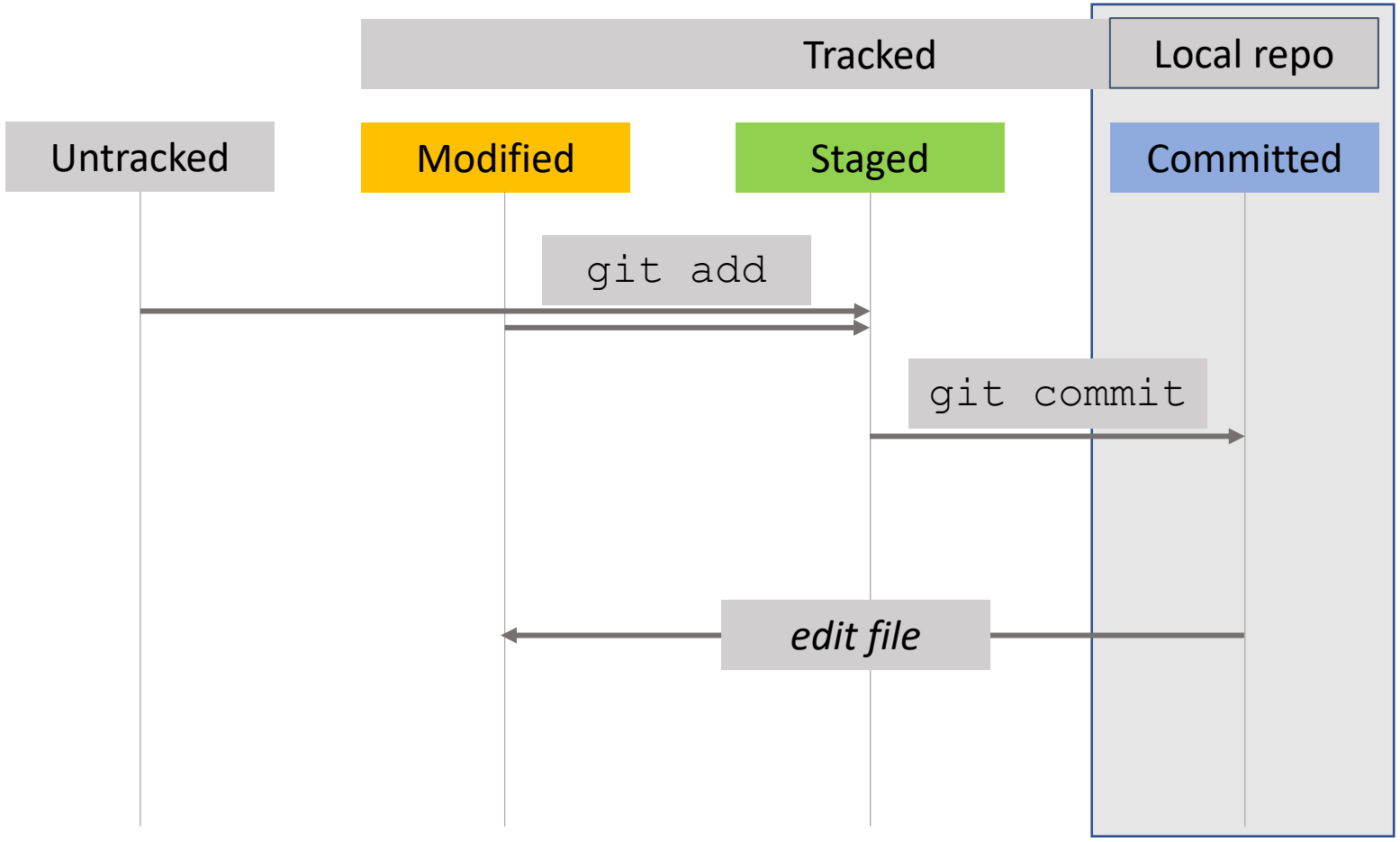
```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git log
```

```
git log <file>
:: show commit history
[for file]
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git log
commit 01b729dee53142b87c85a6802f5f4cc3df38c9a2 (HEAD -> main)
Author: Jennifer Graham <jennifer.graham@sams.ac.uk>
Date:   Mon Mar 30 16:48:14 2026 +0100

    Fixing typo

commit e2c5dde1359e5a03477dc8c30fb2e424c879bcf4
Author: Jennifer Graham <jennifer.graham@sams.ac.uk>
Date:   Mon Mar 30 16:23:56 2026 +0100

    Adding README file

commit d63f8999c47f376c62ca293caedd439697b2cd9c
Author: Jennifer Graham <jennifer.graham@sams.ac.uk>
Date:   Mon Mar 30 16:16:23 2026 +0100

    Adding first file to repository

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
git log <file>
:: show commit history
[for file]
```

```
git init              :: turn current directory into git repository.
git add <file>        :: stage the file for commit i.e., track changes.
git commit            :: confirm changes with message/explanation.


git status            :: show which files have been tracked or modified.
git diff <file>       :: show difference between current file and last commit.
git log <file>        :: show commit history [for file]
```


Tracked
Local repo
Untracked
Modified
Staged
Committed
git add
git commit
edit file

# Aside: repository etiquette - I

- **Do not try to track all your files within a single repository!**
  - *Do not turn your OneDrive or Home into a repository!*

- **Give repositories a <u>meaningful name</u>.**
  - *Think before you create and name your repositories.*
  - Choosing meaningful and consistent names will help both yourself and others.
    - Consider how many repositories could be called “Data_Processing” or “python_scripts”?
    - **Think**: what data is being processed and why? What project are you working on?

- If working on a collaborative project, **discuss with others** to determine what code needs to be included and where, to avoid duplication.

# Aside: repository etiquette - II

- **Git & GitHub are primarily for code/methods, NOT for data storage.**
  - Large data files waste space in the repository, and changes can’t be tracked anyway!
  - However, paths/descriptions of data required should be included, either in the code or associated documentation (e.g., README files).
    - Example data files could be useful/necessary when sharing code.
  - Also, for *small* text or csv data files, you may still find git & GitHub useful for version control.

# Try it yourself… [15 min]

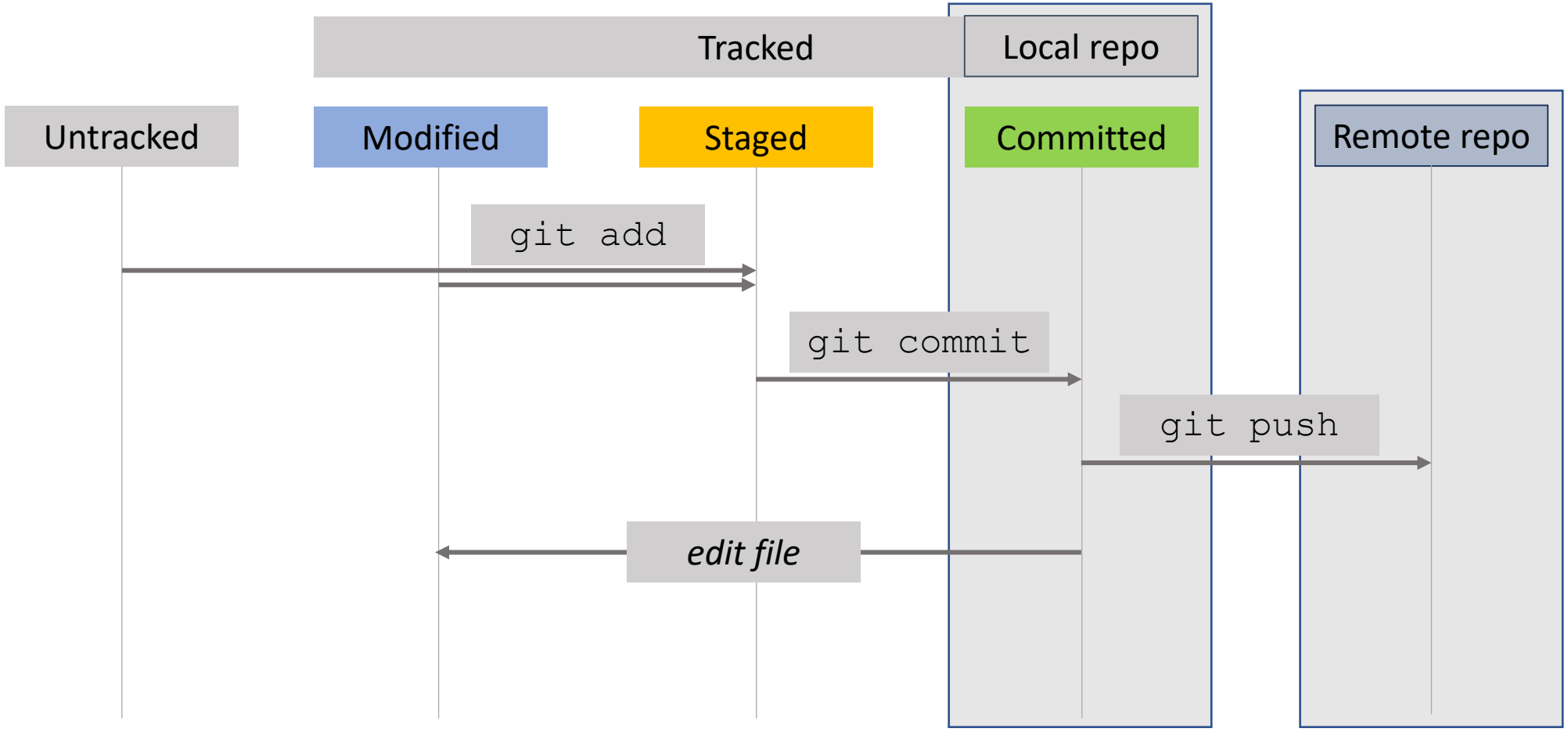
1. Create your own repository locally:
   - Create a new directory, e.g., named “My_First_Repo”.
   - Use “`git init`” to turn it into a repository.
2. Follow the basic workflow to add a simple txt file, make some edits, and then commit changes as you go…

```
git init          :: turn current directory into git repository.
git add <file>    :: stage the file for commit i.e. track changes.
git commit        :: confirm changes with message/explanation.

git status        :: show which files have been tracked or modified.
git diff <file>:: show difference between current file and last commit.
git log <file>    :: show commit history [for file]
```

# Syncing/publishing changes

- Nothing will be visible on GitHub until you “push” your commits.

- You don’t need to “push” every commit (they are still recorded) but remember to do so regularly.
  - Make changes visible for others to see & use.
  - Back-up your code.


Tracked
Local repo
Untracked
Modified
Staged
Committed
Remote repo
git add
git commit
git push
edit file

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git push
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>


JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

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github.com

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Sign in

Sign up

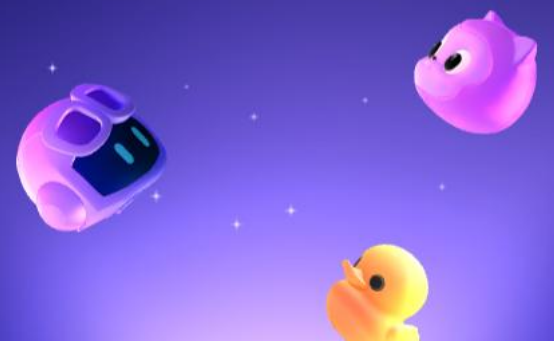
# Build and ship software on a single, collaborative platform

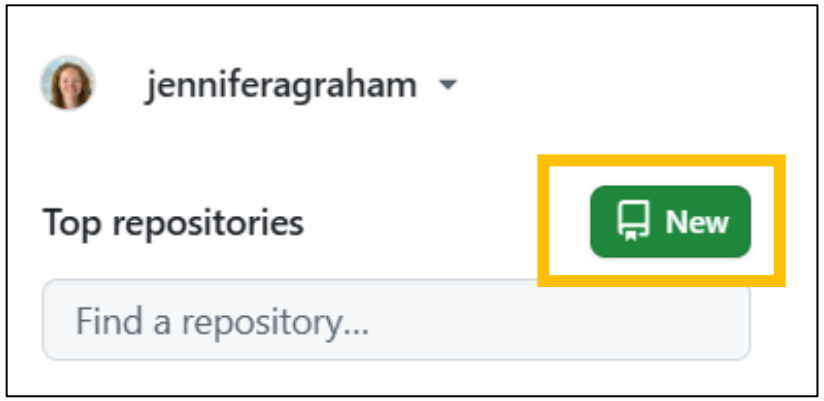
Join the world's most widely adopted AI-powered developer platform.

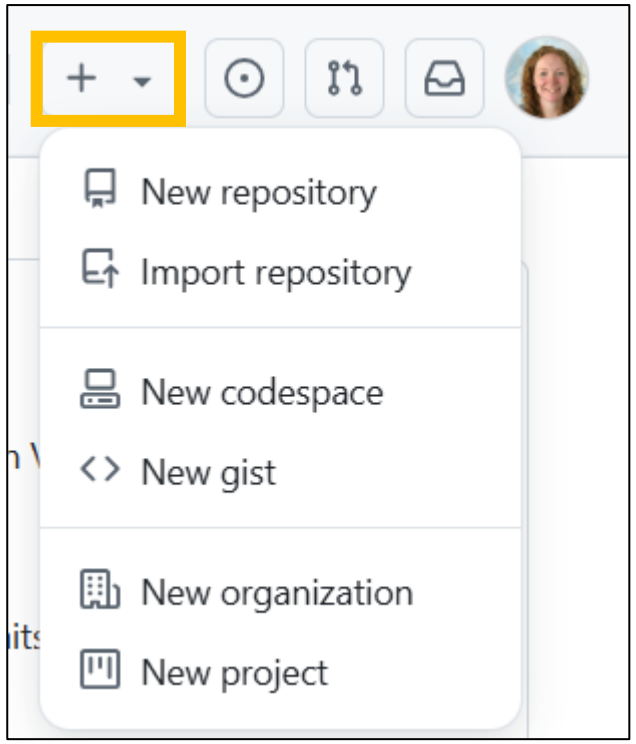
Enter your email

Sign up for GitHub

Try GitHub Copilot






Options to create your new repository.

New repository

https://github.com/new

New repository

Type / to search

## Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.

*Required fields are marked with an asterisk (*).*

1 **General**

Owner *

jenniferagraham

Repository name *

Great repository names are short and memorable. How about didactic-memory?

Description

0 / 350 characters

2 **Configuration**

**Choose visibility ***

Choose who can see and commit to this repository

Public

**Add README**

READMEs can be used as longer descriptions. About READMEs

Off

**Add .gitignore**

.gitignore tells git which files not to track. About ignoring files

No .gitignore

**Add license**

Licenses explain how others can use your code. About licenses

No license

Create repository

NB. If member of an "Organisation", repository can be created/owned there or in personal account.

New repository

https://github.com/new

New repository

Type / to search

## Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.

*Required fields are marked with an asterisk (*).*

### 1 General

Owner *

jenniferagraham /

Repository name *

hello_world

hello_world is available.

Great repository names are short and memorable. How about didactic-memory?

Description

0 / 350 characters

### 2 Configuration

**Choose visibility ***

Choose who can see and commit to this repository

Public

- Public
  Anyone on the internet can see this repository. You choose who can commit.
- Private
  You choose who can see and commit to this repository.

**Add README**

READMEs can be used as longer descriptions. About READMEs

**Add .gitignore**

.gitignore tells git which files not to track. About ignoring files

**Add license**

Licenses explain how others can use your code. About licenses

No license

Create repository

NB. GitHub's default is public, but you may want to ensure control over visibility.

New repository

https://github.com/new

New repository

Type / to search

## Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.

*Required fields are marked with an asterisk (*).*

### 1 General

Owner *

jenniferagraham /

Repository name *

hello_world

hello_world is available.

Great repository names are short and memorable. How about didactic-memory?

Description

0 / 350 characters

### 2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

**DO NOT TICK OR ADD anything in this section**

Add README

READMEs can be used as longer descriptions. About READMEs

Off

Add .gitignore

.gitignore tells git which files not to track. About ignoring files

No .gitignore

Add license

Licenses explain how others can use your code. About licenses

No license

Create repository

jenniferagraham/hello_world

https://github.com/jenniferagraham/hello_world

jenniferagraham / hello_world

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

# hello_world

Public

Pin | Watch 0 | Fork 0 | Star 0

### Set up GitHub Copilot

Use GitHub's AI pair programmer to autocomplete suggestions as you code.

Get started with GitHub Copilot

### Add collaborators to this repository

Search for people using their GitHub username or email address.

Invite collaborators

## Quick setup — if you've done this kind of thing before

Set up in Desktop or HTTPS SSH https://github.com/jenniferagraham/hello_world.git

Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.

## ...or create a new repository on the command line

```
echo "# hello_world" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/jenniferagraham/hello_world.git
git push -u origin main
```

## ...or push an existing repository from the command line

```
git remote add origin https://github.com/jenniferagraham/hello_world.git
git branch -M main
git push -u origin main
```

ProTip! Use the URL for this page when adding GitHub as a remote.

```
$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git remote add origin https://github.com/jenniferagraham/hello_world.git
git branch -M main
git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 14 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 907 bytes | 907.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/jenniferagraham/hello_world.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

Use commands shown on GitHub to set up link and push to remote repository.

jenniferagraham/hello_world

https://github.com/jenniferagraham/hello_world

jenniferagraham / hello_world

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

# hello_world

Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

jenniferagraham Fixing typo 01b729d · 1 hour ago 3 Commits

| | | |
|---|---|---|
| README.md | Adding README file | 2 hours ago |
| hello_world.py | Fixing typo | 1 hour ago |

## README

This is my first repository.

## About

*No description, website, or topics provided.*

Readme
Activity
0 stars
0 watching
0 forks

## Releases

No releases published
Create a new release

## Packages

No packages published
Publish your first package

## Contributors 1

jenniferagraham Jennifer Graham

## Languages

Python 100.0%

## Suggested workflows

Based on your tech stack

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git diff README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git
touches it
diff --git a/README.md b/README.md
index eb6d976..0d08e3c 100644
--- a/README.md
+++ b/README.md
@@ -1 +1 @@
-This is my first repository.
+This is Jenny's first repository.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git
touches it

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git commit
hint: Waiting for your editor to close the file... unix2dos: converting file
C:/Users/sa04jg/OneDrive - SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to DOS format...
dos2unix: converting file C:/Users/sa04jg/OneDrive -
SAMS/GitHub/Repos/hello_world/.git/COMMIT_EDITMSG to Unix format...
[main fe290b3] Specify Jenny's repository
 1 file changed, 1 insertion(+), 1 deletion(-)

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 334 bytes | 334.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jenniferagraham/hello_world.git
   01b729d..fe290b3  main -> main

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

jenniferagraham/hello_world

https://github.com/jenniferagraham/hello_world

jenniferagraham / hello_world

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

# hello_world

Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

jenniferagraham Specify Jenny's repository fe290b3 · 2 minutes ago 4 Commits

| | | |
|---|---|---|
| README.md | Specify Jenny's repository | 2 minutes ago |
| hello_world.py | Fixing typo | 1 hour ago |

## README

This is Jenny's first repository.

## About

*No description, website, or topics provided.*

Readme

Activity

0 stars

0 watching

0 forks

## Releases

No releases published

Create a new release

## Packages

No packages published

Publish your first package

## Contributors 1

jenniferagraham Jennifer Graham

## Languages

Python 100.0%

## Suggested workflows

Based on your tech stack

Commits · jenniferagraham/hello_

https://github.com/jenniferagraham/hello_world/commits/main/

jenniferagraham / hello_world

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

# Commits

main

All users | All time

Commits on Mar 30, 2026

**Specify Jenny's repository**
jenniferagraham committed 3 minutes ago
fe290b3

**Fixing typo**
jenniferagraham committed 1 hour ago
01b729d

**Adding README file**
jenniferagraham committed 2 hours ago
e2c5dde

**Adding first file to repository**
jenniferagraham committed 2 hours ago
d63f899

```
git init              :: turn current directory into git repository.
git add <file>        :: stage the file for commit i.e., track changes.
git commit            :: confirm changes with message/explanation.
git push              :: publish/back-up changes on GitHub (remote server).


git status            :: show which files have been tracked or modified.
git diff <file>       :: show difference between current file and last commit.
git log <file>        :: show commit history [for file]
```

# Try this yourself… [15 min]

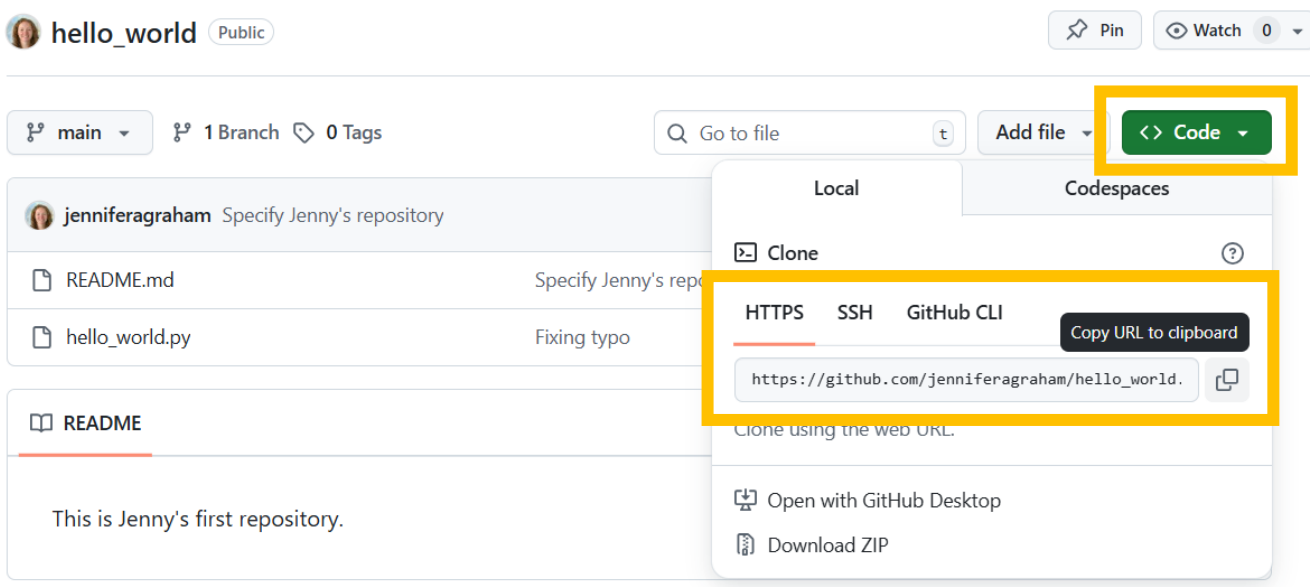
- Push your repository to GitHub, and view changes.
  - Create an ***empty*** repository on GitHub (with the name of your local repository).
  - Use the commands provided to push your local repository to the GitHub site (C&P).

```
git init          :: turn current directory into git repository.
git add <file>    :: stage the file for commit i.e., track changes.
git commit        :: confirm changes with message/explanation.
git push          :: publish/back-up changes on GitHub (remote server).

git status        :: show which files have been tracked or modified.
git diff <file>:: show difference between current file and last commit.
git log <file>    :: show commit history [for file]
```

# Already created a repository on GitHub?

- Use `git clone` to make a local copy.




```
git clone <copied-url>
```

download into new local repository

# [fetch & merge], or pull

- If you make changes directly on GitHub, or once you start collaborating, you may see an error message when you try to push…
- To bring your local repository up to date with remote changes:

```
git fetch        :: brings in information on changes from the remote
git merge        :: merge those changes into your repository

--
```

```
git pull         :: short cut to fetch and merge in one step (if confident
                    that there are no conflicts)
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git push
To https://github.com/jenniferagraham/hello_world.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/jenniferagraham/hello_world.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$
```

jenniferagraham/hello_world

https://github.com/jenniferagraham/hello_world/tree/main

jenniferagraham / hello_world

Type / to search

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# hello_world

Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

jenniferagraham Add title 2d39a54 · 4 minutes ago 5 Commits

| | | |
|---|---|---|
| README.md | Add title | 4 minutes ago |
| hello_world.py | Fixing typo | 2 hours ago |

README

# Hello World

This is Jenny's first repository.

## About

*No description, website, or topics provided.*

Readme

Activity

0 stars

0 watching

0 forks

## Releases

No releases published

Create a new release

## Packages

No packages published

Publish your first package

## Contributors 1

jenniferagraham Jennifer Graham

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git push
To https://github.com/jenniferagraham/hello_world.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/jenniferagraham/hello_world.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$  git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 979 bytes | 65.00 KiB/s, done.
From https://github.com/jenniferagraham/hello_world
   fe290b3..2d39a54  main       -> origin/main
Updating fe290b3..2d39a54
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
```

git pull

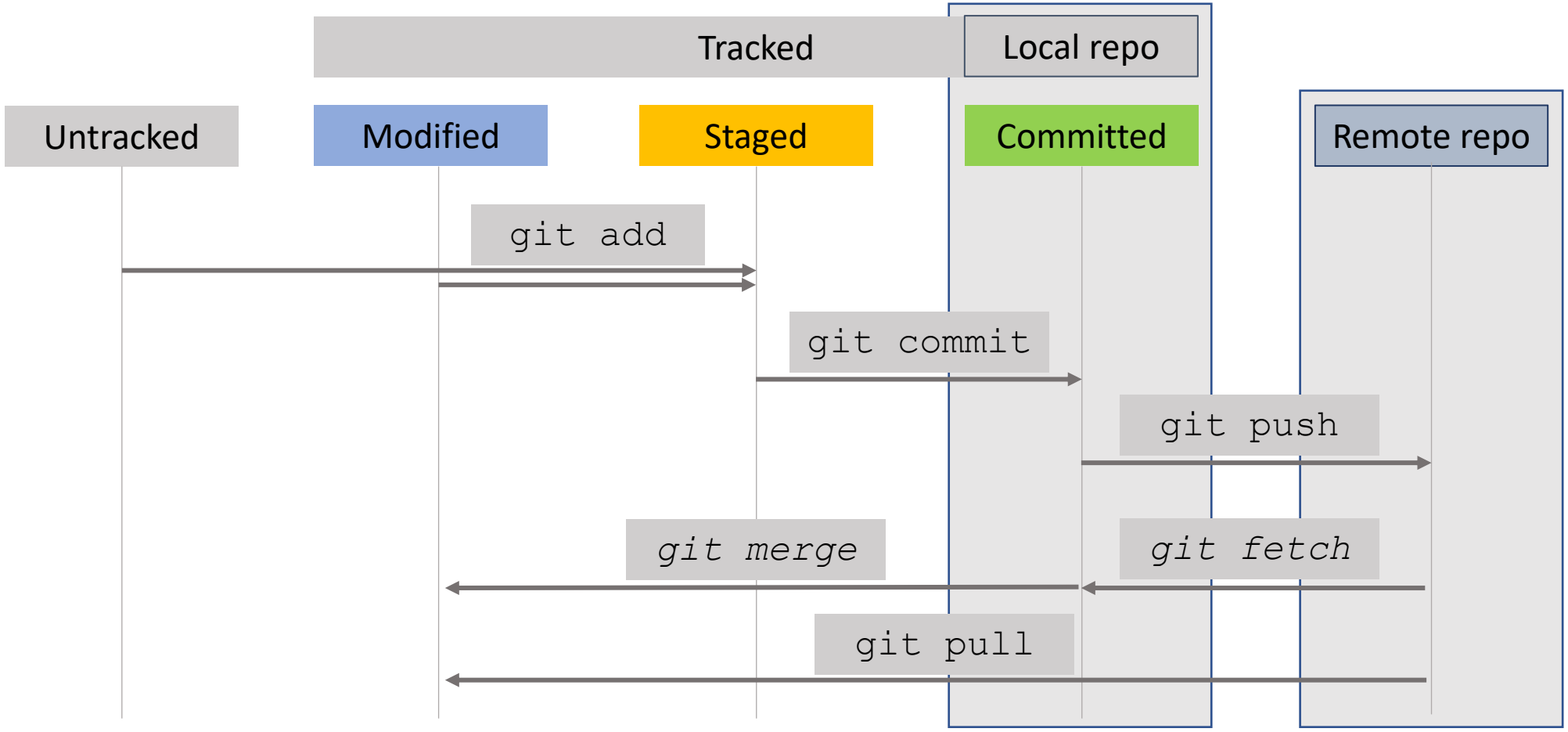
:: fetch updates from remote and merge in one step

# Try it yourself… [10 min]

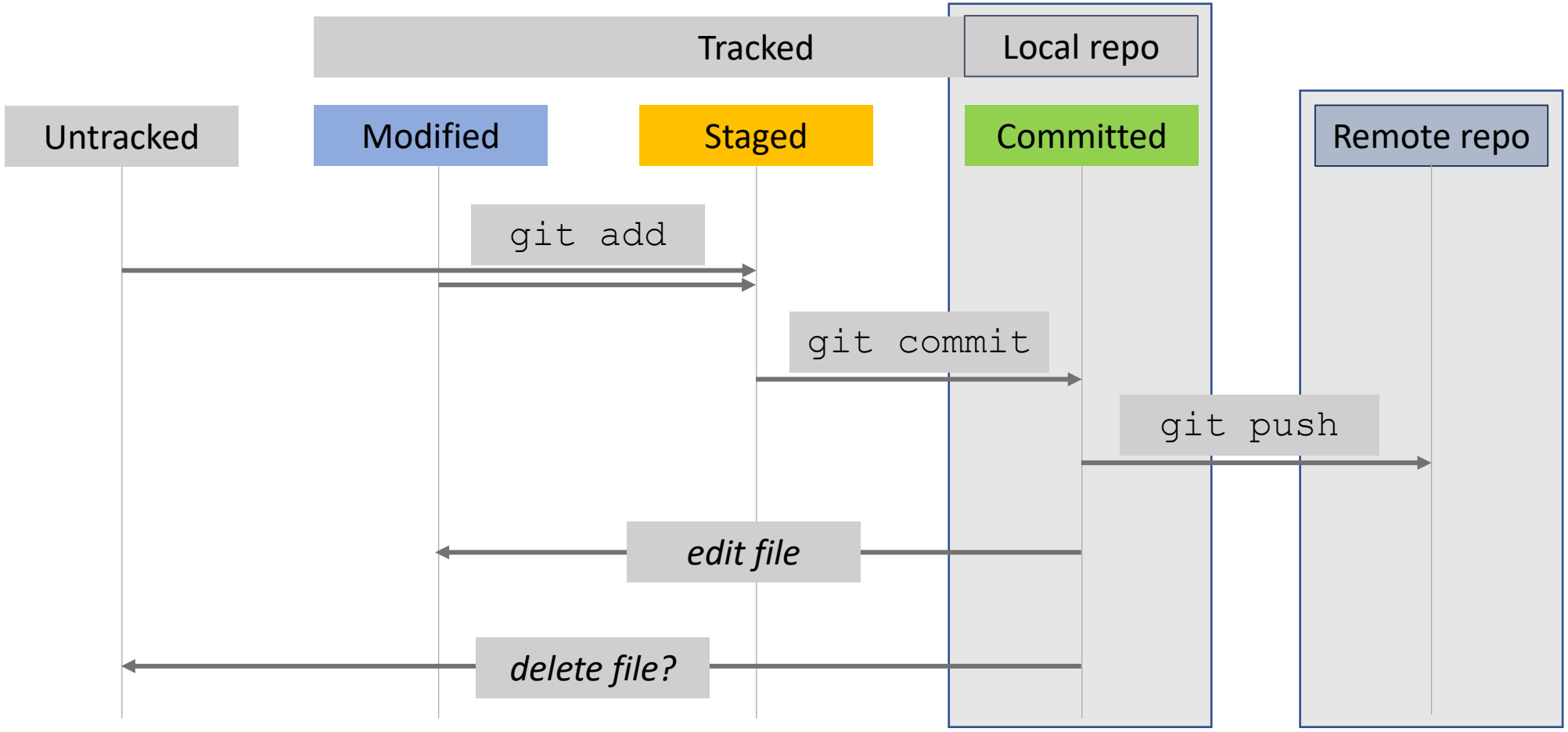
- Edit your repository on GitHub and then pull changes.
  - Does anything break if you try to update the remote and local at same time?

```
git add <file> :: stage the file for commit i.e. track changes.
git commit     :: confirm changes with message/explanation.

git push       :: publish/back-up changes on GitHub (remote server).

git pull       :: retrieve information on remote changes
                  and merge into local repository.
                  [i.e., fetch & merge in one step]

---

git clone <copied-url>   :: download remote into new local repository
```

```
git clone <copied-url>    :: download into new local repository


git pull                  :: retrieve information on remote changes
                                and merge into local repository.
                                [i.e., fetch & merge in one step]
```


Tracked
Local repo
Untracked
Modified
Staged
Committed
Remote repo
git add
git commit
git push
git merge
git fetch
git pull

# More useful tips…


Tracked
Local repo
Untracked
Modified
Staged
Committed
Remote repo
git add
git commit
git push
edit file
delete file?

# Further useful commands…

```
git rm <file>                :: delete file and remove from repository.
git rm --cached <file>       :: delete file from repository, but keep
                                local copy.

git mv <file> <new_name>     :: rename file, and keep tracking.

git remote -v                :: check url for tracked remote repository

git <command> --help         :: open help documentation for any command!
```

# Summary: Terminology

- Repository: Folder containing collection of files to be stored/tracked (any sort of text/code).
- Local repository: Repository being tracked on your local machine (i.e. desktop, hpc).
- Remote repository: a separate copy of your repository stored in a remote location e.g. on GitHub.

- Add: Tells git what files to track / stage files for commit.
- Commit: Saves a snapshot of files, with explanation for any changes.
- Push: Updates from your local machine synced to the remote location.

- Branch: Repositories can be branched into parallel copies, to safely test large developments to the code.
  - The default/central branch is named the “main” (or “master”) branch.
  - This central branch should always contain a working version of the code.
- Head: Latest commit made, tip of the branch in your local repository (current working revision)

# Contents of `.git` folder

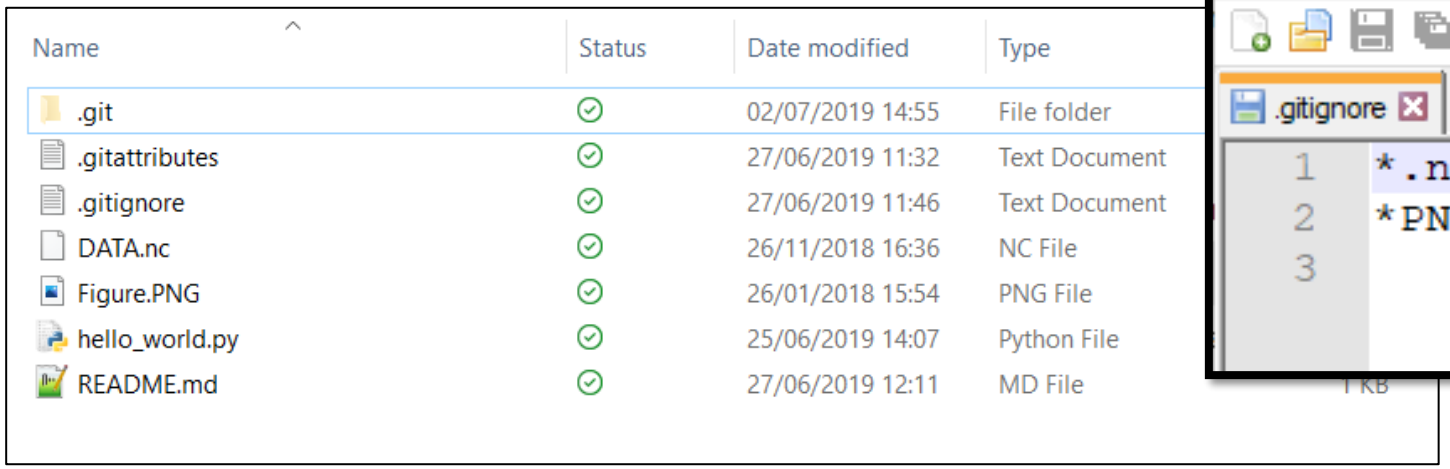
- When a repository is created, this creates a .git folder in the directory.
- .git contains e.g.:

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls -a
./  ../  .git/  README.md  hello_world.py

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ ls .git
COMMIT_EDITMSG  FETCH_HEAD  HEAD  ORIG_HEAD  config  description  hooks/  index  info/  logs/  objects/  refs/
```

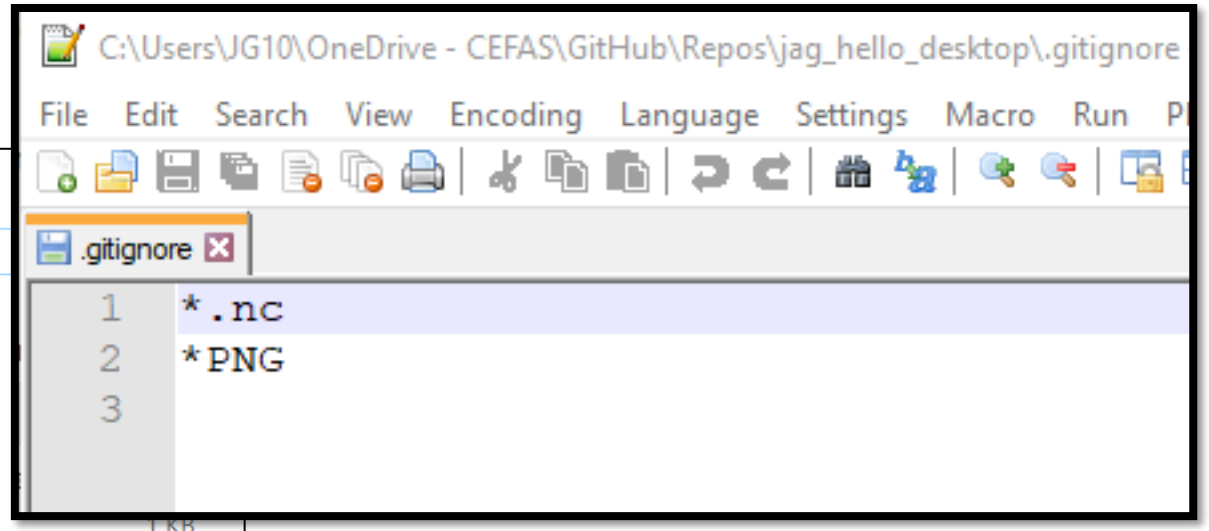
- What are these files?
  - They contain everything that git needs to know about the repository!
  - Whenever you run a command, it will use and/or edit files here.

# .gitignore

- Tell git what to ignore!
- Can ignore any file, directory, or file extensions…
  - e.g. executables, log files, data, figures… (any secrets)
  - Standard examples are provided on GitHub.
- Simple example file:



| Name | Status | Date modified | Type |
|---|---|---|---|
| .git | ✓ | 02/07/2019 14:55 | File folder |
| .gitattributes | ✓ | 27/06/2019 11:32 | Text Document |
| .gitignore | ✓ | 27/06/2019 11:46 | Text Document |
| DATA.nc | ✓ | 26/11/2018 16:36 | NC File |
| Figure.PNG | ✓ | 26/01/2018 15:54 | PNG File |
| hello_world.py | ✓ | 25/06/2019 14:07 | Python File |
| README.md | ✓ | 27/06/2019 12:11 | MD File |



# git config

- Check global username and email settings: `git config --global -l`

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git config --global -l
core.editor=notepad
user.name=Jennifer Graham
user.email=jennifer.graham@sams.ac.uk
```

- Change default editor?

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git config -get core.editor
vi

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git config --global core.editor notepad

JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git config --get core.editor
notepad
```

# git config

- Check local repository settings e.g. path to remote repository

```
git config --local -l
```

```
JG@SAMS MINGW64 ~/OneDrive/GitHub/Repos/hello_world (main)
$ git config --local -l
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/jenniferagraham/hello_world.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
```

# GitHub settings

- Global settings
  - Your account details, notifications, organisation links, etc.
- Repository settings
  - Access, features, visibility, etc.

Your profile

github.com/settings/profile

Settings

Type / to search

Jennifer Graham (jenniferagraham)
Your personal account Switch settings context

Go to your personal profile

- Public profile
- Account
- Appearance
- Accessibility
- Notifications

Access

- Billing and plans
- Emails
- Password and authentication
- Sessions
- SSH and GPG keys
- Organizations
- Enterprises
- Moderation

Code, planning, and automation

- Repositories
- Codespaces
- Packages
- Copilot
- Pages
- Saved replies

Security

- Code security and analysis

# Public profile

**Name**

Jennifer Graham

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

**Public email**

Select a verified email to display

You have set your email address to private. To toggle email privacy, go to email settings and uncheck "Keep my email address private."

**Bio**

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

**Pronouns**

Don't specify

**URL**

**ORCID iD**

ORCID provides a persistent identifier - an ORCID iD - that distinguishes you from other researchers. Learn more at ORCID.org.

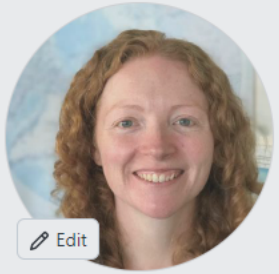
Connect your ORCID iD

**Social accounts**

Link to social profile

Link to social profile

**Profile picture**



Edit

jenniferagraham
Jennifer Graham

- Set status
- Your profile
- Add account
- Your repositories
- Your projects
- Your Copilot
- Your organizations
- Your enterprises
- Your stars
- Your sponsors
- Your gists
- Upgrade
- Try Enterprise Free
- Feature preview
- Settings
- GitHub Support
- GitHub Community
- Sign out

https://github.com/settings/profile

General

Access
Collaborators

Code and automation
Branches
Tags
Rules
Actions
Webhooks
Codespaces
Pages

Security
Code security and analysis
Deploy keys
Secrets and variables

Integrations
GitHub Apps
Email notifications

# General

Repository name

hello_world Rename

☐ **Template repository**
Template repositories let users generate new repositories with the same directory structure and files. Learn more about template repositories.

☐ **Require contributors to sign off on web-based commits**
Enabling this setting will require contributors to sign off on commits made through GitHub's web interface. Signing off is a way for contributors to affirm that their commit complies with the repository's terms, commonly the Developer Certificate of Origin (DCO). Learn more about signing off on commits.

## Default branch

The default branch is considered the "base" branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.

main

## Features

☐ **Wikis**
Wikis host documentation for your repository.

**Upgrade or make this repository public to enable Wikis**
GitHub Wikis is a simple way to let others contribute content. Any GitHub user can create and edit pages to use for documentation, examples, support, or anything you wish.

Upgrade Learn more about wikis

# Adding documentation

- README
  - Automatically the front page of your repository.
  - Markdown formatting can be used
- Wiki
  - Broader information on use of repository.
  - Only available within public repositories or organisations.
  - e.g. https://github.com/CefasRepRes/Git_Training/wiki/
- Issues
  - Flagging development stages or known problems.
- Projects
  - Organise work/issues on repositories?

- Some repository tracking info only available within organisations or paid accounts
  - More info here: https://github.com/pricing

# Aside: main vs master

- GitHub now uses “main” as the default central branch name (rather than master)
  - In 2020, other software companies announced similar changes.
- If you’re working with older software, or creating repositories locally with an older version of git, you may find that “master” is still the default branch name.
  - Your default will depend on your local git settings.
  - You may remember this being an option while installing git on your PC?
- There are a few different ways you can update this…

# How to change the “central” branch name?

**If you have just setup a new repository on GitHub, but don’t want to rename anything locally:**

```
git remote add origin <GitHub_Path>
git push -u origin master
```

**If you have just setup a new repository on GitHub, and want to rename your current local branch to main:**

```
git remote add origin <GitHub_Path>
# Rename the current branch to main
git branch -M main
git push -u origin main
```

**If you rename an existing repository on GitHub, and want to update your local repository**

```
# Rename the master branch to main
git branch -m master main
git fetch origin
git branch -u origin/main main
git remote set-head origin -a
```

**NB. GitHub will give tips for the above when creating a new repository.**